Henry Vu

469-649-6967 | Vuducdung28@gmail.com | linkedin.com/in/henry-vu27 | github.com/HenryVu27 | henryvu.io

Experience

Machine Learning Engineer Aug 2025 – Present

eXRealityAI | Python, SQL, vision-language models, ARKit, Unity (C#), Swift *Dallas, TX*

- Lead engineer on SmartDVI, a Meta Quest 3 vehicle-inspection app in production at repair shops, directing its development and owning its AI layer from photo grading to on-device agents, VIN reading and voice.
- Built its **on-device agent framework**, where agents fire on inspection events such as a photo, voice note or OBD session, are bound to checklist items by shop admins as auto-apply or suggest-only, and never overwrite a technician's entry.
- Wrote **vision agents** that read odometers, tread gauges, dipsticks, coolant levels, plates and service stickers from photos, writing a value only above a confidence floor and otherwise offering it to the technician.
- Built the evaluation behind its photo grading from graded production inspections, comparing model and prompt configurations to choose the one the backend adopted.
- Tested the model's stated confidence as a gate for grades and found it carried no usable signal, so AI grades stay suggestions the technician confirms.
- Built VIN reading as a vision cascade that ranks a burst of frames by sharpness, decodes the barcode, falls back to a vision model that flags uncertain characters, and repairs them with the check digit.
- On XR Auto Service, a guided-repair headset app, built the **knowledge-graph retrieval layer** behind its repair assistant, filtered to the exact vehicle on the lift and scored on expert-written cases on every change.

Computer Vision Engineer Feb 2025 – Present

ThorMed Innovation | PyTorch, U-Net, SimSiam, quantization *Dallas, TX*

- Led domain-aligned transfer learning for an NIH-funded bladder segmentation project, pretraining U-Net encoders on 9,204 thyroid and breast ultrasound images, supervised and with SimSiam, to reach **96% Dice**.
- Quantized the models to 4-bit weights for portable ultrasound devices, where domain-pretrained encoders kept **75 to 93% Dice** while ImageNet-initialized ones fell to 11 to 71%.

Undergraduate Researcher Apr 2022 – May 2024

University of Alberta, Amii and SODALab *Edmonton, AB*

- Implemented primal-dual online algorithms that use machine-learned predictions, and benchmarked multi-armed bandit algorithms (UCB, Exp3, Thompson sampling) across stochastic, adversarial and restless settings. [REPO]

Projects

Medical Imaging | *PyTorch, OpenCV, Weights & Biases*

- Ran joint-embedding predictive architecture (JEPA) pretraining and segmentation fine-tuning on bladder ultrasound in a collaborator's research codebase, a second self-supervised family alongside the SimSiam work above.
- Generated bladder segmentation masks with the SAMUS ultrasound foundation model and a refinement step. [REPO]
- Trained U-Net and attention U-Net models on the Kvasir-SEG polyp and BUSI breast ultrasound datasets. [REPO]

ADHDAgent, guardrailed agentic RAG | *LangGraph, Qdrant, ONNX, React 19*

- Built a **LangGraph ReAct agent** between two guardrail gates, with hybrid Qdrant retrieval, a local cross-encoder reranker and an offline evaluation package scored by LLM judges. [REPO]

DatDai, deployed legal RAG assistant | *FastAPI, Gemini, Qdrant, Langfuse, Cloud Run*

- Vietnamese land-law assistant, evaluated on live traffic by replaying production traces through an LLM judge. [REPO]

Education

University of Texas at Dallas Richardson, TX

M.Sc. Computer Science, Intelligent Systems (GPA 4.0/4.0) *Sep 2024 – Summer 2027 (expected)*

University of Alberta Edmonton, AB

B.Sc. Computing Science with Honors (GPA 3.9/4.0, *Summa cum laude*) *Sep 2019 – May 2024*

Technical Skills

Languages: Python, SQL, C#, Swift
Machine Learning: PyTorch, Hugging Face, OpenCV, ONNX, Weights & Biases
Models: U-Net, Segment Anything (SAM), YOLO, Whisper, vision-language models
LLMs and Agents: LangGraph, LangSmith, Langfuse, RAG, Qdrant, FAISS
Platforms: Docker, Google Cloud, PostgreSQL, GitHub Actions, Unity, OpenXR, ARKit